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## Lab Report 03: Edge Detection Techniques and Their Impact on Classification Performance

ISIC 9-Class Skin Cancer Dataset Comparative Study

### 1. Introduction

Edge detection is a foundational classical image processing technique designed to locate sharp spatial intensity discontinuities corresponding to object boundaries and surface textures. While edge maps provide compact structural representations of images, their utility in complex multi-class diagnostic tasks remains a topic of investigation. This report evaluates traditional edge detection operators (Sobel, Prewitt, Laplacian, LoG, and Canny), analyzes their noise sensitivity and parameter configurations, and compares classification performance across raw, pre-filtered, and edge-only representations using the ISIC 9-Class Skin Lesion Dataset.

### 2. Methodology

- **Edge Detection Operators:** Implemented first-order gradient operators (Sobel, Prewitt), second-order derivative operators (Laplacian, Laplacian of Gaussian), and multi-stage edge detection (Canny) across representative lesion classes.
- **Noise Analysis & Pre-filtering:** Artificial Gaussian noise and Salt-and-Pepper noise were added to test image variants. The responses of edge operators were evaluated on unfiltered noisy images as well as images pre-smoothed with Gaussian and Median filters.
- **Canny Parameter Optimization:** We benchmarked four Canny parameter configurations combining different low/high hysteresis thresholds (30/100, 50/150, 100/200) and Gaussian smoothing kernel sizes (3x3, 5x5).
- **Cross-Lab Model Evaluation:** Models were trained and evaluated using identical 80/20 stratified dataset splits across three distinct image representations: Set A (Raw Images - Lab 01), Set B (Filtered Images - Lab 02), and Set C (Edge Maps - Lab 03).

### 3. Experimental Setup

- **Dataset:** ISIC 9-Class Skin Cancer Dataset (nodoubttome/skin-cancer9-classesisic)
- **Image Resolution:** 224 x 224 pixels normalized to standard ImageNet parameters.
- **Classical Classifiers:** Support Vector Machine (RBF-SVM), Random Forest (100 trees), and K-Nearest Neighbors (K=5) trained on extracted ResNet18 feature embeddings.

- Deep Learning Models: ResNet18 and ResNet50 fine-tuned using Adam optimizer (learning rate =  $1e-4$ ) and Cross-Entropy loss.
- Hardware Environment: PyTorch 2.x with CUDA acceleration.

#### 4. Results Tables

Table 1. Effect of Noise and Preprocessing on Edge Detection

| Edge Detector | Input Image | Noise Type    | Preprocessing   | Edge Quality | Noise Sensitivity | Observations                                                           |
|---------------|-------------|---------------|-----------------|--------------|-------------------|------------------------------------------------------------------------|
| Sobel         | Original    | None          | None            | High         | Low               | Clear lesion boundaries with fine texture details.                     |
| Sobel         | Noisy       | Gaussian      | None            | Poor         | High              | High false edges across homogenous skin regions.                       |
| Sobel         | Noisy       | Salt & Pepper | None            | Very Poor    | Very High         | Impulse noise artifacts produce severe isolated edge spikes.           |
| Sobel         | Noisy       | Gaussian      | Gaussian Filter | Moderate     | Low               | Noise suppressed effectively; edges slightly blurred.                  |
| Sobel         | Noisy       | Salt & Pepper | Median Filter   | Good         | Low               | Median filter completely eliminates salt-and-pepper noise artifacts.   |
| Prewitt       | Original    | None          | None            | High         | Low               | Similar boundary response to Sobel but slightly weaker diagonal edges. |
| Laplacian     | Original    | None          | None            | Moderate     | Very High         | Second-order derivative                                                |

|       |          |               |                    |           |          |                                                                              |
|-------|----------|---------------|--------------------|-----------|----------|------------------------------------------------------------------------------|
|       |          |               |                    |           |          | accentuates minor intensity changes.                                         |
| LoG   | Noisy    | Gaussian      | Gaussian Filter    | Moderate  | Moderate | Pre-smoothing suppresses high-frequency noise before derivative computation. |
| Canny | Original | None          | Built-in smoothing | Very High | Low      | Thin, continuous, 1-pixel wide edge contours of lesion boundaries.           |
| Canny | Noisy    | Gaussian      | Gaussian Filter    | Good      | Low      | Non-maximum suppression and hysteresis preserve primary boundary contours.   |
| Canny | Noisy    | Salt & Pepper | Median Filter      | Good      | Low      | Median filter removes impulse spots, preventing false hysteresis triggers.   |

Table 2. Canny Parameter Analysis

| Configuration | Low Threshold | High Threshold | Kernel Size | Edge Quality        | Number of Detected Edges | Observation                                                    |
|---------------|---------------|----------------|-------------|---------------------|--------------------------|----------------------------------------------------------------|
| Canny-1       | 30            | 100            | 3x3         | High Detail / Noisy | 4,812                    | Captures fine internal textures but includes background noise. |

|         |     |     |     |                        |       |                                                                       |
|---------|-----|-----|-----|------------------------|-------|-----------------------------------------------------------------------|
| Canny-2 | 50  | 150 | 3x3 | Optimal /<br>Balanced  | 2,945 | Best trade-off: clear continuous lesion boundary without clutter.     |
| Canny-3 | 100 | 200 | 3x3 | Sparse /<br>Fragmented | 1,120 | High threshold rejects minor details, causing broken boundary lines.  |
| Canny-4 | 50  | 150 | 5x5 | Smooth /<br>General    | 2,310 | Larger 5x5 kernel smooths out fine skin textures before thresholding. |

Table 3. Cross-Lab Classification Performance Comparison

| Model /<br>Classifier     | Accuracy Raw<br>(Lab 1)<br>(%) | Accuracy<br>Filtered<br>(Lab 2)<br>(%) | Accuracy Edge<br>(Lab 3)<br>(%) | Precision (%) | Recall (%) | F1-Score (%) | Training Time (s) | Inference Time (ms) |
|---------------------------|--------------------------------|----------------------------------------|---------------------------------|---------------|------------|--------------|-------------------|---------------------|
| SVM                       | 60.17                          | 59.40                                  | 41.20                           | 59.49         | 60.17      | 57.81        | 12.45             | 1.85                |
| Random Forest             | 50.42                          | 49.80                                  | 33.10                           | 49.82         | 50.42      | 45.91        | 8.30              | 0.95                |
| KNN                       | 48.73                          | 47.90                                  | 30.50                           | 50.15         | 48.73      | 47.61        | 0.05              | 4.20                |
| CNN Model 1<br>(ResNet18) | 68.01                          | 66.80                                  | 42.90                           | 67.46         | 68.01      | 67.22        | 45.30             | 17.26               |
| CNN Model 2<br>(ResNet50) | 68.86                          | 66.10                                  | 43.20                           | 67.14         | 68.86      | 65.16        | 95.10             | 18.14               |

## 5. Discussion & Answers to Task Questions

- **Q1 (Edge Detection & Noise):** Un-smoothed Laplacian and Sobel detectors exhibited extreme noise sensitivity. Second-order derivatives compute spatial intensity differences quadratically, causing noise spikes to generate widespread false edge artifacts across uniform regions.
- **Q2 (Effect of Pre-Filtering):** Pre-smoothing with Gaussian filters suppressed continuous noise, while Median filtering effectively eliminated Salt-and-Pepper impulse artifacts, preventing false edge triggers.
- **Q3 (Canny Threshold Analysis):** Lower thresholds (30/100) produced dense edge maps capturing internal lesion textures alongside background noise. High thresholds (100/200) rejected minor details, producing fragmented, broken boundary contours. The configuration Low=50, High=150 provided the optimal balance.
- **Q4 (Impact of Edge Maps):** Using edge-only images (Set C) caused a drastic drop in classification accuracy (~25% decline) across all models compared to raw images.
- **Q5 (Information Loss):** Converting dermoscopy images to binary edge maps removes chromatic information (color hue/saturation), pigment network intensity, shading, and internal texture details, which are critical clinical indicators in skin lesion diagnosis.
- **Q6 (Classical vs. Deep Learned Features):** CNNs automatically learn task-specific directional edge kernels in early convolutional layers during backpropagation. Allowing the network to learn these features dynamically produces rich multi-scale representations superior to static handcrafted edge maps.
- **Q7 (Optimal Dataset Representation):** Raw images (Set A) produced the highest classification accuracy across all models. Retaining full textural and chromatic information provides the optimal feature space for automated diagnosis.

## 6. Conclusion

This laboratory demonstrates that while classical edge detection operators effectively isolate structural boundaries, stripping away color distribution and internal texture severely degrades machine learning classification performance. Deep learning models operate best on raw, full-spectrum image data where early convolutional layers can automatically extract dynamic, task-optimized feature hierarchies.